



English, Economics, Politics, and Philosophy

Unit 2 Topics in Business

Lesson 2.4: Determination of Interest and Rationality (Part I)

Pertinent Topics:

- Determination of Interest Rate
- Rationality
- Prisoner's Dilemma
- Individuality vs Collective optimum

Success Criteria:

- I will understand the meaning of rationality
- I will learn how the interest rate is determined

MINDS ON

A difficult question: suppose the Central Bank of Taiwan prints money constantly, would hundred dollars today value the same as hundred-dollar next year? Try to connect the idea of scarcity to this question. Also, what happens to the opportunity cost of money if interest rate changes according to this scenario. (You will have to decide whether interest rate decreases/increases as a result of increased money printing).



The central bank shouldn't print loads of money.

Because the price (nominal i.e., the number) will be crazy.

If the bank dumps money, then the interest will be trashed.

According to the law of scarcity, when we have a lot of money, money is not scarce. Money is going to be not as valuable, prices become higher. Hundred dollar today will not value the same: hundred dollar next year will be less valuable than hundred dollar today

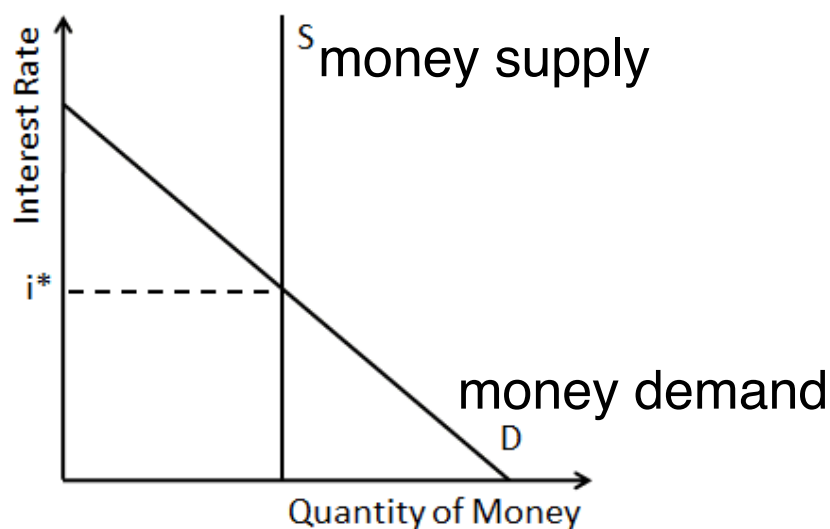
The opportunity cost of money:

if I had invested/saved it at the bank, what would I have earned

I. Determination of Interest Rate

Last time we talked about the money's value of **counterfactual usage, i.e., if had I invested/saved it at the bank, what would I have earned**. This is essentially attributable to the fact that the bank lends out your deposited money to other people as loans and earned the principles. However, what we did not explore was the determination of interest rate. How exactly is interest rate set?

There are in fact two channels through which interest rate is determined: loanable bond market and supply-demand money market. Today we will talk about supply-demand money market. Now, intuitively we know that the market should be constituted by Money Supply/Money Demand (just as how a typical supply demand curve works):



- MS is made by two things: circular currency (CC) demand deposit (DD). CC is basically the cash that we are holding. DD is the money we save in the bank.
- MD on the other hand indicates how **desirable is money to us at the current moment**. As mentioned, money, if not used, has an opportunity cost equals to its interest rate. The higher the interest rate (i.e., price of money), the lower our desire to have money. Because when interest rate is high, asset price becomes cheaper, (e.g., stock price becomes lower), people tend to invest more. So that in the future when the interest rate is lower, the asset price becomes higher (stock price becomes higher) (a profit can be made).

present value calculation

Text

GE	221.90	-3.30	-1.5%	114.81	111.93	687,279
IBM	112.56	-2.57	-2.2%	3879.65	3696.4	228,906
MSFT	3716.78	-148.50	-3.8%	298.69	282.04	578,141
AMZN	282.04	-17.08	-5.7%	96.75	93.57	207,773
GOOGL	93.57	-3.19	-3.3%	91.17	89.47	356,383
FB	89.93	-1.03	-1.1%	1077.38	1019.23	336,667
DIS	1019.23	-57.62	-5.4%	1503.79	1457.54	8,278,841
BA	1457.54	-40.92	-2.7%	72.96	71.44	891,639
WMT	71.54	-1.43	-2.0%	3417.92	3320.78	245,470
PG	3320.78	-105.40	-3.1%	137.21	133.27	270,076
MRK	133.27	-4.14	-3.0%	5774.98	5648.86	1,263,941
UNH	5648.86	-128.45	-2.2%	2622.58	2596.33	1,263,941
CVX	2596.33	-115.56	-4.4%			

At the moment, you would only need a broad picture for these ideas. In tutorials later we will learn about stock market.

rationality: reasonable (理性) vs emotional/sentimental (感性)

II. Rationality

Think about it what does it mean to be rational? Does it mean

egoism? Or does it mean the pursuit of self-interest? Or, more

self-centred, the pursuit of best outcome profit motive
generously, the pursuit of best interest for all people?

Q: what is rationality to you?

In economics, rationality is interpreted as the pursuit of self-interest. But why is that? Can we ever justify this seemingly egocentric pursuit? Adam Smith says yes; when everyone realizes their full potentials, then what happens to the society?

(the idea of Invisible Hand kicks in)

Pareto: collective optimum--rational just is to have
the best outcome for all people

John Nash: non-regret equilibrium—rational just is
to be non-regretful

Melissa: rationality is a protection for myself and other people; reason can be used as a means to protect you and your friends. E.g., if milk is spilled mom is angry, you have to use your reason to articulate properly to explain yourself.

Assignment:

Before we end today's tutorial here is some foods for thought before our next session. Suppose there are two players, player A and player B, both are charged by prosecutors. Now, the prosecutors only have some evidences against both of them, but to charge them fully, some further evidence has to suffice.



Both players know the followings:

1. Whoever provides evidence (defect) will be freed if the other person remains silent (cooperate). This silent person serves for three years.
2. If both provides evidence, then both serve for 2 years.
3. If both cooperate, then both serve for 1 year.

Player A		Player B
Actions	Cooperate	Defect
Cooperate	(1, 1)	(3, 0)
Defect	(0, 3)	(2, 2)

Question: Imagine you are player A, what should you do?

Note: this is a difficult question as well, but I would encourage you to at least attempt reasoning a strategy. Some theories regarding rational decision will be covered next time.